

Baran Polat

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Education

Mersin University

Bachelor of Science in Computer Engineering
Thesis [Note: Optional]
Relevant Coursework: Data Structures, Algorithms, Machine Learning, Database Systems.

Mersin,Turkey
June,2025

Experience

ETERNA TECHNOLOGY

Data Scientist & Developer

Mersin,Turkey
Nov 2024, Present

- Developing Python scripts for automated data extraction (ETL) and preprocessing pipelines to reduce manual data entry time by 40%.
- Optimizing machine learning model performance by conducting hyperparameter tuning and feature engineering on real-world datasets.

MINDSANE

Machine Learning Engineer

Istanbul,Turkey
Aug 2025 – Sep 2025

- Implemented Reinforcement Learning algorithms for stock market forecasting simulation, achieving a 15% improvement in prediction accuracy.
- Collaborated with senior engineers to deploy ML models (YOLO) for computer vision tasks in a production-like environment.

PRODENSO INFORMATION TECHNOLOGIES

Software Engineer

Istanbul,Turkey
Nov 2024 – Mar 2025

- Maintained company database systems (PostgreSQL/MySQL), ensuring 99.9% uptime through regular health checks and monitoring scripts.
- Generated automated system health reports using Python, enabling the IT team to identify potential server issues proactively.

PROJECTS

Child Dental Assistant

Python, YOLOv8

- Secured research funding from TÜBİTAK to develop an AI-powered mobile diagnostics tool for pediatric dental health.
- Trained and fine-tuned a custom YOLOv8 model on a specialized dental dataset, achieving high precision in detecting caries and anomalies.
- Designed an automated image processing pipeline to analyze user-uploaded photos and generate instant health reports.

Financial News Sentiment & Market Analysis

Python, NLP, Web Scraping, Pandas

- Engineered a data extraction pipeline using Selenium/BeautifulSoup to scrape over 15,000 financial news articles.
- Implemented Natural Language Processing (NLP) techniques to classify market sentiment (Positive/Negative/Neutral), creating a dataset for algorithmic trading signals.
- Visualized correlation between news sentiment and market trends to aid in financial decision-making.

Real-Time License Plate Recognition System

Python, YOLOv5, OpenCV

- Developed a high-performance vehicle detection system capable of recognizing license plates in real-time video streams.
- Optimized the YOLOv5 inference process to ensure low-latency performance on standard hardware environments.
- Integrated OCR (Optical Character Recognition) to convert detected plate images into digital text format for database logging.

Precious Metal Price Forecasting (LSTM)

TensorFlow, Keras, Time-Series Analysis

- Designed a Deep Learning model using Long Short-Term Memory (LSTM) networks to predict Gold and Silver price movements.
- Processed historical financial data and integrated technical indicators (RSI, MACD) to improve prediction accuracy by reducing noise.

Clustering & Customer Segmentation Analysis

Scikit-learn, PCA, Unsupervised Learning

- Applied advanced clustering algorithms (K-Means, DBSCAN) to segment complex datasets for pattern recognition.
- Utilized Principal Component Analysis (PCA) for dimensionality reduction, enabling clearer visualization of high-dimensional data.

Skills & Interests

Languages: Python, SQL, C++,Flutter.

Data Science & AI: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, YOLO, NLP, OpenCV.

Tools & Platforms: Git, Docker, VS Code, Linux.

Languages: Turkish (Native), English (Professional).

Interests: Algorithmic Trading, Home Automation (IoT & Arduino), Kaggle Competitions.